Benchmark of comparative single-cell analysis approaches

Gregor Habitzreither Sebastian Fay Marie Schygulla

Structure Teamproject 2026

Supervised by Prof. Dr. Manfred Claassen and Pouria Laghaee

11. September 2026

Benchmark of comparative single-cell analysis approaches

The Data

from Horowitz et al. (2013)

Provided donors	20
CMV+/CMV-	9/11
Analysis markers	37
Available gated NK cells	261,593
Balanced analysis sample	40,000

The context

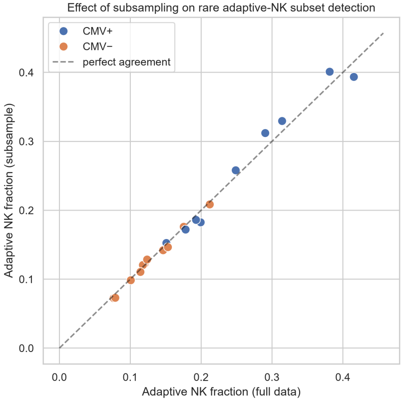
Mass cytometry (CyTOF): simultaneous protein-marker measurements on individual cells. **Natural Killer (NK) cells**: immune cells with a high cell diversity

From some donors a subpopulation of NK cells shows a characteristic expression of proteins typical after a CMV infection, which was given in the data

CMV labels describe donors; they do not identify infected individual cells.

Horowitz et al., Sci. Transl. Med. 2013; Arvaniti & Claassen, Nat. Commun. 2017. Provided 20-donor subset

Does the balanced subsample preserve the selected phenotype?

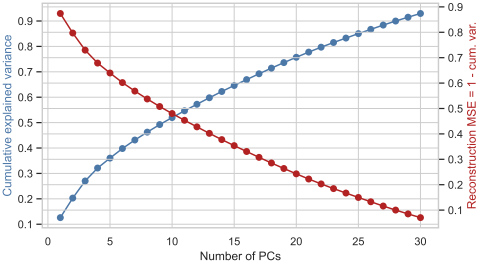


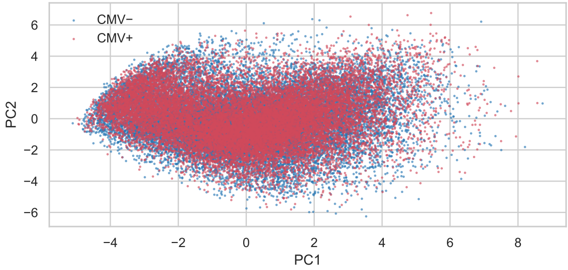
Compare randomly choosen **2,000 cells per donor** with all available gated NK cells from that donor.

NKG2C$^+$/CD57$^+$ is a marker phenotype associated with **memory-like ("adaptive") NK cells** after CMV exposure.

Positivity: two-component GMM crossover per marker on all 261,593 cells. This check concerns one approximate phenotype, not every rare population.

PCA



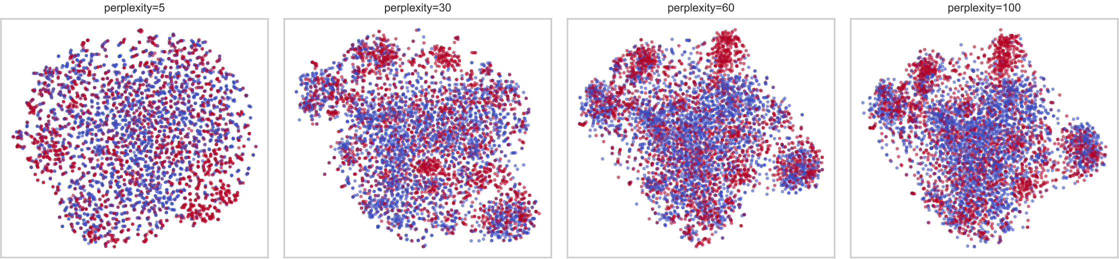


Retention choice: 29 PCs reach the 90% threshold and become the input to t-SNE/UMAP.

29 PCs retain 91.50%, the plotted two retain only 20.29%

40,000 cells, arcsinh + z-scoring. Spectrum verified from the same 37-marker covariance matrix

t-SNE: changing perplexity changes the neighbourhood scale



Input: 5,000 cells and 29 PCs reduce sweep time and memory.

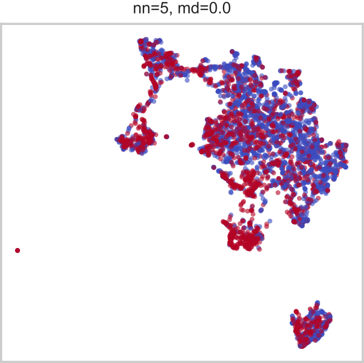
The top-scoring of 5, 15, 30, 40, 50, 60, 70 and 100. **Selected: 60**

Perplexity	Trustworthiness
50	0.92597
60	0.92605
100	0.92580

T: penalises false low-dimensional neighbours by their high-dimensional ranks.

Selected panels; colours indicate donor CMV status.

UMAP

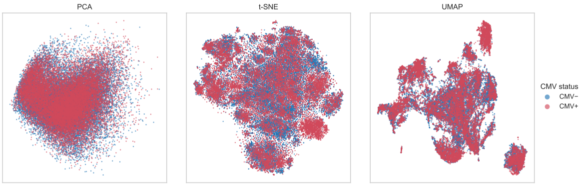


$n_{\text{neighbors}}$	min_dist	$T(15)$
5	0.0	0.890139
5	0.1	0.879145
30	0.0	0.877151

Selected: **5 neighbours, min_dist = 0.0**.

Four corner settings from the 16-panel saved grid, colours indicate CMV status. Top three rows ranked by saved Trustworthiness on the same 5,000 cells and 29 PCs.

Local preservation and global distances favour different maps



Embedding	$T(15)\uparrow$	$R(15)\uparrow$	$r_d\uparrow$	$S_{\mathrm{CMV}}\uparrow$
PCA	0.7258	0.0072	**0.6204**	0.0016
t-SNE	**0.9612**	**0.2232**	0.4868	0.0104
UMAP	0.9009	0.1126	0.3954	**0.0178**

Local structure

T: **Trustworthiness** – false neighbours, ranked.

R: **kNN preservation** – exact overlap; small values are normal at $n = 40{,}000$ (chance level ≈ 0.0004).

t-SNE wins both: it directly optimises neighbour probabilities.

Global / label structure

r_d: **Pearson correlation** of all pairwise distances. *PCA wins: a linear projection built to preserve overall variance/distance.*

S_{CMV}: **Silhouette score** - separation by donor CMV label – near zero for all three maps, as expected: only a minority subset of each CMV+ donor's cells is actually altered.

t-SNE leads locally; PCA leads in distance correlation.

40,000 cells; 29 PCs; k=15. Distance correlation: fixed 2,000-cell sample.

Which maps are most similar?

Pair	$R(15)\uparrow$	$M^2\downarrow$
PCA vs t-SNE	0.00585	**0.45189**
PCA vs UMAP	0.00410	0.73390
t-SNE vs UMAP	**0.12532**	0.72369

R values look small in absolute terms, but sit far above the ~ 0.0004 level expected by chance at $n = 40{,}000$, $k = 15$.

M^2 – Procrustes disparity

Align two maps by translation, rotation/reflection and scaling; measure the remaining mismatch. Lower = closer overall shape after alignment.

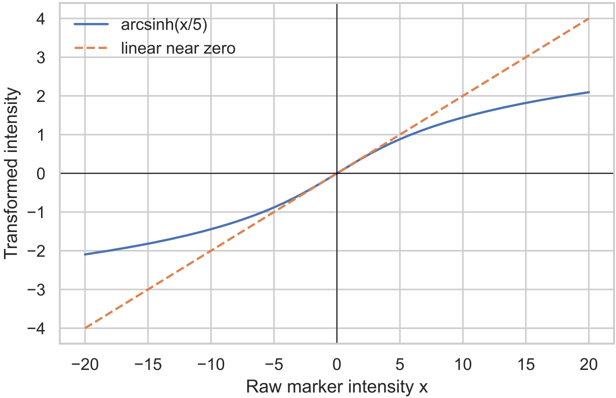
Local neighbours (R): t-SNE and UMAP are the most similar pair.

Aligned global shape (M^2): PCA and t-SNE are the most similar pair.

R and M^2 disagree because they ask different questions: exact shared neighbours vs. overall shape after optimal alignment

All three pairs evaluated on the same 40,000 cells in the same order. R(15) corrected from saved coordinates; SciPy Procrustes values independently reproduced. Neither metric identifies biologically correct cell types.

Backup: Data Transformation and z-scoring



The arcsinh transformation

$$a_{ij} = \operatorname{asinh}(x_{ij}/5)$$

Approximately linear near zero; compresses large magnitudes and accepts negative background values.

Marker-wise z-scoring

$$z_{ij} = \frac{a_{ij} - \bar{a}_j}{s_j}$$

Prevents large marker scales from dominating distances. All 37 markers pass the variance (0.05) filter.

cofactor 5; pooled scaling across the balanced exploratory sample. Curve is a mathematical illustration.

Backup: how t-SNE turns neighbours into a map

High-D affinity (probability that j is a neighbour of i):

$$p_{ij} \propto \exp(-\|x_i - x_j\|^2 / 2\sigma_i^2)$$

Low-D affinity (heavy-tailed, avoids "crowding"):

$$q_{ij} \propto (1 + \|y_i - y_j\|^2)^{-1}$$

Optimisation moves points in 2D until q_{ij} matches p_{ij} as closely as possible (minimises $\mathrm{KL}(P\|Q)$).

σ_i is tuned per point so its affinities imply the chosen perplexity.

perplexity	swept (5–100); effective neighbourl size – see main slide.
init = pca	starts from the PCA layout instead of dom noise: more reproducible, avoids local optima.
learning rate = auto	optimiser step size; scikit-learn scal with sample size.
early exagg. = 12	temporarily inflates high-D affinities on, so initial clusters separate before positioning.
angle = 0.5	Barnes–Hut speed/accuracy trade-of numerical setting, not a property of data.
max_iter = 750	fixed iteration budget – not itself a co gence check (see caveat below).

van der Maaten & Hinton (2008); scikit-learn TSNE documentation. A fixed iteration count alone does not confirm convergence. Defaults refer to the documented scikit-learn 1.9 environment.

Backup: kNN preservation (R)

Let $N_i^X(k)$, $N_i^Y(k)$ be the k nearest neighbours of cell i in the high-D reference X and the embedding Y (excluding itself).

$$R(k) = \frac{1}{n} \sum_{i=1}^{n} \frac{|N_i^X(k) \cap N_i^Y(k)|}{k}$$

Symmetric; counts exact shared identities, not just "close enough."

What k controls. How many neighbours count as "local." Small k is a strict test of fine-grained ordering; large k is a looser test that tolerates more reshuffling and starts to resemble cluster-membership agreement rather than precise neighbour identity.

Why $k = 15$ here. Fixed throughout to match the neighbourhood scale explored in the UMAP sweep (`n_neighbors` 5–50) and sits within the commonly used 10–30 range for trustworthiness-style scores at this dataset size.

Example: ten shared neighbours out of fifteen contribute $10/15$.

Venna & Kaski (2001); scikit-learn NearestNeighbors documentation. Full-sample scores average over cells; equal cells per donor give equal donor weight.

Backup: Trustworthiness (T)

Using the same $k = 15$ and the same neighbour sets $N_i^X(k)$, $N_i^Y(k)$ as R:

$$T(k) = 1 - \frac{2}{nk(2n - 3k - 1)} \sum_i \sum_{j \in N_i^Y(k) \setminus N_i^X(k)} (r_X(i,j) - k)$$

Penalises *false* low-D neighbours (points that look close in the embedding but weren't close in the reference) according to how far out of the true top-k they actually ranked. Unlike R, direction matters here: T only punishes false positives, not missed true neighbours.

Example: a false neighbour originally ranked 16th (just outside the true top-15) contributes only $(16 - 15) = 1$ to the penalty; one ranked 500th contributes 485 – a much larger embedding error is penalised much more heavily.

Venna & Kaski (2001); scikit-learn trustworthiness documentation. Here $k = 15$, matching R.

Backup: how UMAP constructs and embeds a weighted graph

$$w_{j|i} = \exp\left[-\frac{\max(0, d(x_i, x_j) - \rho_i)}{\sigma_i}\right], \qquad w_{ij} = w_{j|i} + w_{i|j} - w_{j|i}w_{i|j}.$$

Offset ρ_i and scale σ_i adapt to local density; directed neighbourhoods are combined into symmetric weights.

$$v_{ij} = (1 + a\|y_i - y_j\|^{2b})^{-1}, \qquad L \sim -\sum_{i<j}\{w_{ij}\log v_{ij} + (1 - w_{ij})\log(1 - v_{ij})\}.$$

Sampled updates balance attraction and repulsion. The parameters a, b depend on min_dist and spread.

Swept	n_neighbors: 5, 15, 30, 50; min_dist: 0, 0.1, 0.3, 0.5.
Fixed/default	2 dimensions, seed 42, Euclidean metric, spectral initialisation, spread 1.

Larger neighbourhoods change the scale of the graph; larger min_dist discourages tight packing. Neither guarantees accurate global distances.

McInnes, Healy & Melville (2018/2020); official UMAP parameter and algorithm documentation. Equations describe the conceptual objective, not every optimisation approximation.

Backup: how UMAP constructs and embeds a weighted graph

Step 1 – build the high-D graph.

$$w_{j|i} = \exp\left[-\frac{\max(0, d(x_i, x_j) - \rho_i)}{\sigma_i}\right], \qquad w_{ij} = w_{j|i} + w_{i|j} - w_{j|i}w_{i|j}.$$

Offset ρ_i and scale σ_i **adapt to local density**; directed neighbourhoods are combined into symmetric edge weights.

Step 2 – fit a low-D layout to match it.

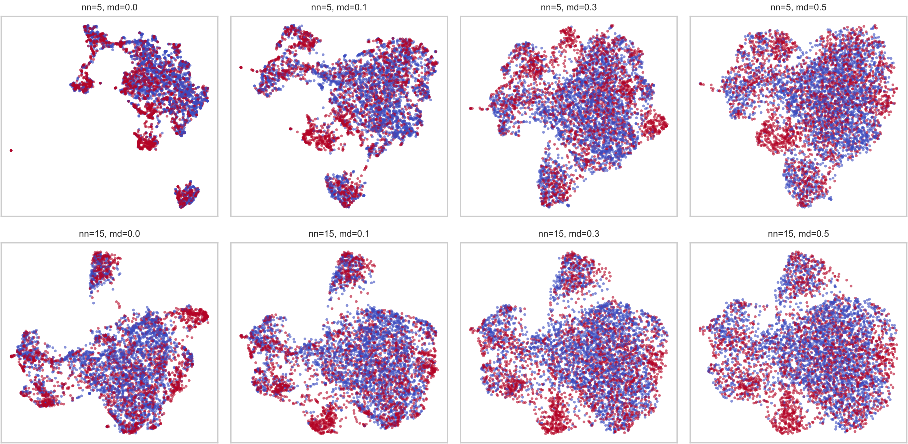
$$v_{ij} = (1 + a\|y_i - y_j\|^{2b})^{-1}, \qquad L \sim -\sum_{i<j}\{w_{ij}\log v_{ij} + (1 - w_{ij})\log(1 - v_{ij})\}.$$

Sampled updates balance **attraction** (true neighbours pulled together) and **repulsion** (non-neighbours pushed apart). a, b are fit automatically from `min_dist`/`spread`, not set directly.

Swept	**n_neighbors**: 5, 15, 30, 50; **min_dist**: 0, 0.1, 0.3, 0.5.
Fixed/default	2 dimensions, seed 42, Euclidean metric, spectral initialisation, spread 1.

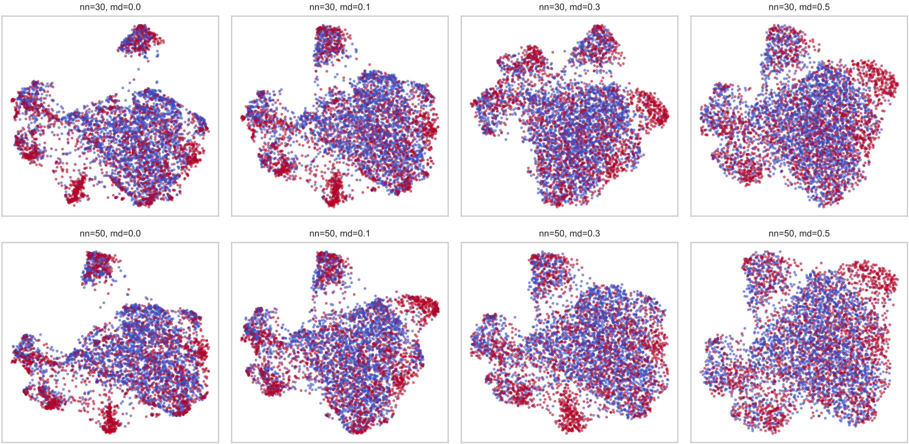
Larger neighbourhoods change the **scale** of the graph; larger `min_dist` discourages tight packing. Neither guarantees accurate global distances.

Backup: UMAP sweep, 5 and 15 neighbours



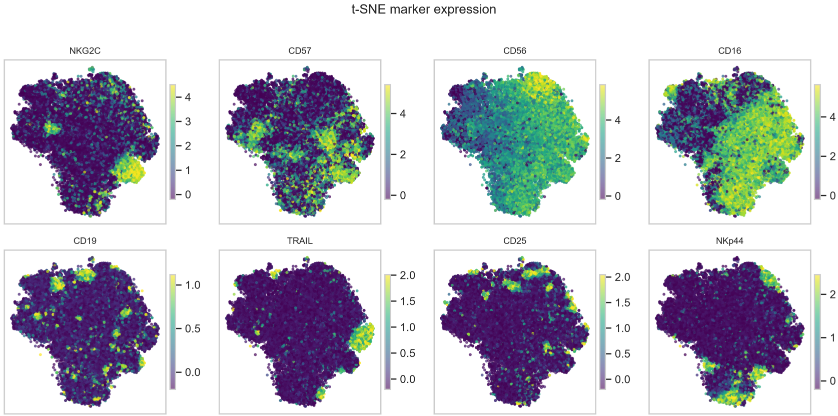
Rows 1–2 of the original 16-panel grid. Same 5,000 cells, 29 PCs, seed 42; Visual changes are parameter effects within one run, not a stability study across seeds.

Backup: UMAP sweep, 30 and 50 neighbours



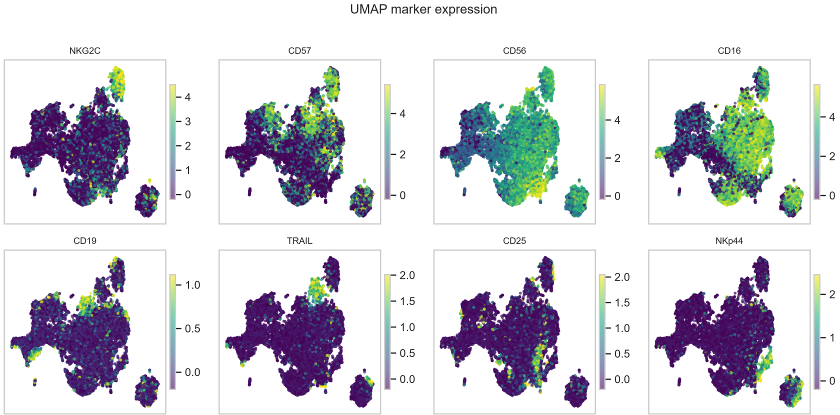
Rows 3–4 of the original 16-panel grid. Same 5,000 cells, 29 PCs, seed 42; Visual changes are parameter effects within one run, not a stability study across seeds.

Backup: the same markers on the t-SNE map



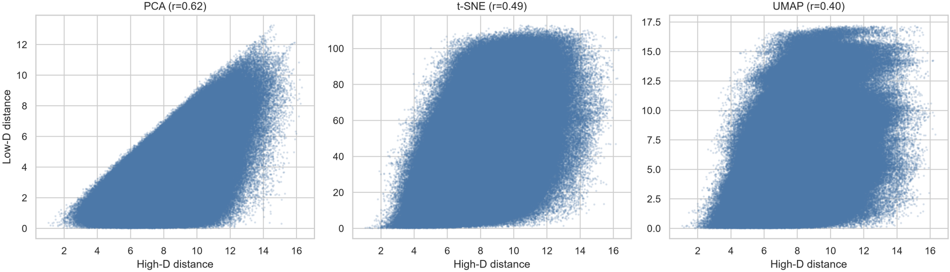
40,000 balanced cells. Arcsinh-transformed marker values; colour limits use the 1st/99th percentiles separately per marker. A bright region does not establish an infected-cell population.

Backup: marker expression on the UMAP map



40,000 balanced cells. Arcsinh-transformed marker values; colour limits use the 1st/99th percentiles separately per marker. A bright region does not establish an infected-cell population.

Backup: Shepard diagram to compare distances



Each point represents one sampled cell pair. The **horizontal axis** is the distance in the 29-dimensional PCA reference space.

The **vertical axis** is the distance in the two-dimensional embedding. Pearson correlation measures agreement in relative distances.

UMAP's loss doesn't optimize for preserving absolute distances at all, and `min_dist`=0 actively encourages tight packing, so many far-apart high-D pairs get compressed into a similar low-D distance ceiling → lowest r_d of the three

Fixed 2,000-cell sample; 29-PC reference space; Pearson correlation of pairwise distances.

Backup: Procrustes aligns the maps before measuring mismatch

Centre and normalise each coordinate matrix to unit Frobenius norm:

$$A_c = \frac{A - \bar{A}}{\|A - \bar{A}\|_F}, \qquad B_c = \frac{B - \bar{B}}{\|B - \bar{B}\|_F}.$$

$$M^2 = \min_{s, Q:\, Q^\top Q = I} \|A_c - sB_c Q\|_F^2.$$

- Rows must correspond to exactly the same cells in the same order.
- Rotation/reflection, translation and uniform scaling do not count as biological differences.
- Lower disparity means a closer global fit after alignment; it does **not** establish local fidelity (whether each point's *immediate* neighbours are correctly placed – that's what T/R measure instead) or class separation.
- Nonlinear deformations can change neighbour identities and global shape differently.

SciPy scipy.spatial.procrustes. With its normalisation and optimal scaling, the scalar disparity is symmetric even though the returned transformed matrices have different normalisation conventions.

Sources

Data and biology

Horowitz et al. (2013). *Genetic and environmental determinants of human NK cell diversity revealed by mass cytometry.* Sci. Transl. Med. 5:208ra145.

Arvaniti & Claassen (2017). *Sensitive detection of rare disease-associated cell subsets via representation learning.* Nat. Commun. 8:14825.

Methods

van der Maaten & Hinton (2008), JMLR 9:2579–2605 (t-SNE).

McInnes, Healy & Melville (2018/2020), arXiv:1802.03426 (UMAP).

Venna & Kaski (2001), ICANN, pp. 485–491 (Trustworthiness).

Official scikit-learn and SciPy documentation; linked references in the detailed report.